GENED 1104: SCIENCE & COOKING MIDTERM 2

Thursday, November 16, 2023, 12-1:15 pm Calculators permitted

Expectations

- **Read the directions carefully.**
- **For problems involving calculation, please show your work.**
- **Academic Integrity:**

  Cheating, no matter how slight, is never acceptable.
  It violates the standards of our community and disrupts learning.
  A higher score is not worth compromising your integrity.

  - I have not consulted anyone or any reference material in completing this exam.
  - I will not discuss exam questions and solutions until the exam is handed back.
  - I understand and agree to abide by this policy:

____
Name (Printed)

____
(Signature)

Section Time/TF (circle):

Tu 3:45 Alex	Tu 6:45 Kate	W 12:45 Caroline	W 3:45 HyoJeon	W 6:45 Blake	Th 9:45 Tim	Th 3:45 Larissa

Grading (to be completed by teaching staff)

Question	Score
1-6	/18
7-9	/16
10-12	/14
13-16	/25
16-21	/27
Total	/100

Multiple Choice Questions: (3 pts each)

1. At what volume fraction does the elastic modulus of an emulsion start increasing?
   a. 50%
   b. 1%
   c. 74%
   d. 64%

2. Which of the following statements is true about diffusion? Select all that apply.
   a. Diffusion is always faster for molecules with a positive or negative charge
   b. The overall direction of diffusion is from areas of high concentration to low
   c. Large molecules take longer steps in a "random walk" motion and therefore diffuse faster
   d. Moving a dish into a hotter environment will speed up diffusion of heat into the food but will have no effect on diffusion of ions

3. Which of the following are true? Select all that apply.
   a. Ostwald ripening happens between two differently sized droplets due to differences in internal pressure.
   b. Creaming happens in substances with different densities.
   c. Laplace pressure describes why there is a difference in pressure between small and large droplets.
   d. Phase inversion occurs when the volume fraction of an emulsion rises above 1.

4. What types of molecules are required for Maillard reactions? Select all that apply.
   a. Fats
   b. Carbohydrates
   c. Salts
   d. Proteins

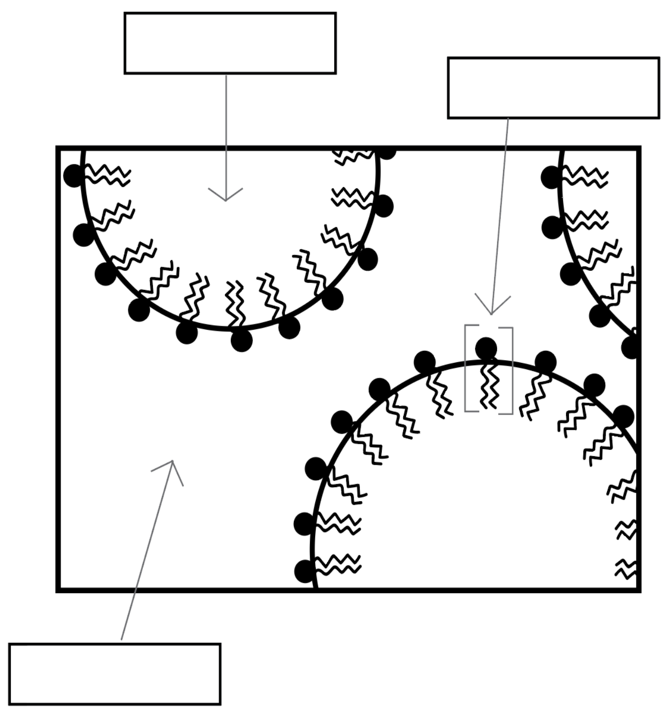
5. Which of the following are true about polymers? Select all that apply.
   a. A polymer's effective volume can be estimated by the principles of a "random walk"
   b. Flour is commonly used in recipes because it is a better thickener than xanthan gum
   c. Polymers that are less likely to entangle are typically better thickening agents
   d. Polymers affect the rate that molecules in a liquid can move around each other

6. Draw lines between each food and what is diffusing into it. Connect all that apply; i.e. each food can have more than one line.

Tuna ceviche	H+ ions
Molten lava cake	Oil
French fries	CO gas
BBQ ribs	Heat

Short Answer Questions:

7. What type of emulsion is represented in the following picture? In each box, write in the depicted component and in 1-2 sentences explain how you reached your conclusion. (6 pts)



8. Why does spherification not work as well with low-pH substances? Explain. (5 pts)

9. Can you have an oil-in-oil emulsion? Explain why or why not. (5 pts)

Hot Chocolate

Head TF Heide visits a local shop where they show her how cocoa beans are fermented, dried and ground to form cocoa powder. She uses this cocoa powder to make a hot chocolate by gently stirring together 10g of cocoa powder, 10g of sugar and 500mL of milk while heating on the stove.

10. Is Heide's hot chocolate a colloid, emulsion and/or foam? Explain why or why not for all three. (6 pts)

11. Heide measures that her recipe makes 510mL of hot chocolate. If cocoa powder has a density of 1.6 g/mL, what is the volume fraction of cocoa powder in Heide's hot chocolate? (4 pts)

12. Heide thinks her hot chocolate could be just a bit thicker. <u>Without adding any ingredients</u>, what could she do to thicken her hot chocolate? Using the volume fraction equation, explain how this would lead to an increase in viscosity. (4 pts)

Steak Dinner

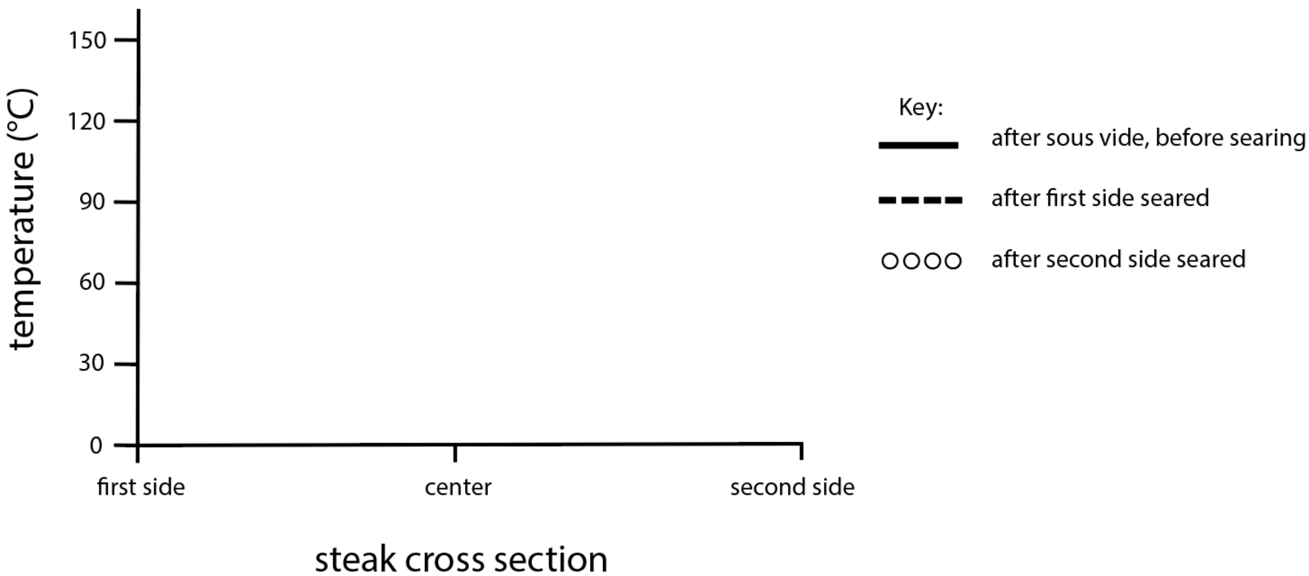
TF Tim wants to make the perfect steak for a dinner party. He decides to cook it using a sous vide water bath set to 55ºC, the temperature of a medium-rare steak. He seals the steak in a plastic bag with some herbs and seasoning, and submerges the steak completely in the water bath to cook.

13. Tim's steak is a rectangular block with dimensions 10cm x 10cm x 6cm. What is the shortest amount of time (in minutes) that Tim needs to leave the steak in the water bath to ensure the center of the steak reaches 55ºC? Note: the diffusion coefficient of heat for beef is 1.3×10^{-3} cm^2/sec. (5 pts)

14. Tim wants his steak to be ready faster, so he decides to make a thinner slice. How thick should he make the steak if he wants the sous vide cooking to take half as long? Assume that the length and width of the cut steak are still both 10 cm as in Problem #13. (5 pts)

15. After his steak has finished in the sous vide bath, Tim quickly sears both the top and bottom sides on high heat just until Maillard reactions occur, about a few minutes each.

    Using the axes provided, draw three temperature profiles for Tim's steak (1) immediately after the sous vide but before searing, (2) immediately after searing the first side and (3) immediately after searing the second side. Please use the key provided for drawing the three different profiles. (9 pts)



16. Tim wants to add more flavor into his steak by marinating it in red wine vinegar, olive oil, and salt in the fridge for 12 hours before repeating the above recipe for cooking. However, he finds that this time the steak is overcooked! Explain why this happened on a molecular level. (6 pts)

Dalgona Coffee

For an extra fancy morning, TF Larissa likes to make “Dalgona coffee”. You make this type of coffee by vigorously whisking equal parts instant coffee (readily dissolvable powdered coffee), hot water, and white sugar until it foams up.

17. One morning Larissa prepared her coffee, but then forgot about it. When she came back after a while the coffee foam had disappeared. Give two reasons why this may have occurred, describing the processes and what causes them. (6 pts)

18. Larissa’s friend suggests she use an electric whisk or blender instead of whisking the coffee by hand. This not only makes it foam more quickly, but the foam is also much stiffer. Explain <u>in terms of droplet size</u> what you think may have changed and why. (6 pts)

19. Larissa whisks 80mL of coffee mixture until the total volume reaches 250mL. Assume the droplet radius for the dispersed phase is 5μm (10^6 μm = 1m) and that the surface tension is 0.02N/m. What is the elastic modulus of the foam? (6 pts)

20. Larissa's friend also suggests adding 1 tsp of milk to try to stabilize the whipped coffee. Larissa suggests adding 1 tsp of egg whites instead. Which method would work better and why? (5 pts)

21. After adding this stabilizer, Larissa calculates that the volume fraction of air in her whipped coffee is 0.60. Would this coffee technically be considered a solid? Would it be considerably viscous? Briefly provide your reasoning. (4 pts)

Equation Sheet

Unit conversions				
N = kg·m/s^2	J = N·m	Pa = N/m^2 = J/m^3	1 L= $1\cdot10^{-3}$m^3	1 ml = 1 cm^3
Kelvin = Celsius+273	Fahrenheit = (Celsius*9/5)+32	Area circle = πr^2	Volume sphere = $(4/3)\pi r^3$	Surface area sphere = $4\pi r^2$

Constants			
Room temp = 23˚C	1Cal = 1000 cal = 4.18 kJ	Gravitational acceleration (g=-9.8 m/s^2)	Avogadro's number ($6.022\cdot10^{23}$ molecules/mol)
Boltzmann constant k_B = ($1.38\cdot10^{-23}$ J/K)	Specific heat water 4.19 J/gK	Latent heat of vaporization for water 2260 J/g	Latent heat of fusion for water 334 J/g
1 mol = 6.022 x 10^{23} molecules	gas constant = 22.4L/mol	D_{heat} in water $1.4\cdot10^{-3}$ cm^2/sec	D_{Ca} $8.0\cdot10^{-10}$ m^2/sec

Equation		Description	Units
$L = \sqrt{4Dt}$	L	Distance of diffusion	m
	D	Diffusion coefficient	m^2/s
	t	Time to diffuse	s
$\phi = \dfrac{V_{dispersed}}{V_{total}}$	ϕ	Volume fraction	-
	$V_{dispersed}$	Volume of dispersed phase	L
	V_{total}	Total volume	L
$E = \dfrac{\sigma}{R}(\phi - \phi_C) \qquad \phi > \phi_C$	E	Elastic modulus	Pa
	σ	Surface tension	N/m
	R	Droplet radius	m
	ϕ	Dispersed phase volume fraction	-
	ϕ_C	Critical volume fraction (0.64 for spheres)	-

Equation		Description	Units
$\mathrm{pH} = -\log_{10}[\mathrm{H}^+]$	$[H^+]$	Concentration of protons (also known as H^+ ions) in solution	mol/L
	pH	A measure of acidity	none
$Q = mc_p\Delta T$	Q	Heat	J
	m	Mass	g
	c_p	Specific heat capacity	J/g K or J/g˚C
	ΔT	Change in temperature	K or ˚C
$Q = mL_v$	Q	Heat	J
$Q = mL_f$	m	Mass	g
	L_v	Latent heat of vaporization	J/g
	L_f	Latent heat of fusion	J/g
$U = Ck_BT$	U	Interaction energy for liquid to gas transitions	J
	C	Constant (1.5 for water)	none
	k_B	Boltzmann constant	J/K
	T	Temperature	K
$E = \dfrac{F/A}{\Delta L / L_0}$	E	Elastic modulus	Pa
	F	Force	N
	A	Area over which force is applied	m^2
	ΔL	Change in length	m
	L_0	Initial length	m
$E = \dfrac{U_{\mathrm{int}}}{l^3} = \dfrac{k_BT}{l^3}$	E	Elastic modulus	Pa
	k_B	Boltzmann constant	J/K
	T	Temperature	K
	l	Average distance between crosslinks	m
$N(t) = N_0e^{kt}$ where	$N(t)$	Number of cells at time t	microbes
$k = \dfrac{\ln 2}{\tau}$	N_0	Number of cells at time 0	microbes
	k	Growth or death rate of cells	s^{-1}
	t	Time elapsed	s
	τ	Doubling time	s